

Sara Colando

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 GitHub

EDUCATION

Ph.D. in Statistics, Carnegie Mellon University, Pittsburgh, PA 2024-2029 (*expected*)

M.S. in Statistics, Carnegie Mellon University, Pittsburgh, PA 2024-2026

B.A. in Mathematics and Philosophy, Pomona College, Claremont, CA 2020-2024

Magna cum laude (GPA: 3.99/4.00) and departmental honors

- Math thesis supervised by Jo Hardin, titled "Selecting ChIP-Seq Normalization Methods from the Perspective of their Technical Conditions."
- Philosophy literature review supervised by Julie Tannenbaum, titled "What are the Key Philosophical Conceptions of Fairness and Justice?" with an emphasis on conceptions relevant to algorithmic fairness.

PUBLICATIONS AND PAPERS

• Peer-reviewed articles

2. Colando, S., Schulz, D., Hardin, J. **Selecting ChIP-Seq Normalization Methods from the Perspective of their Technical Conditions**, *Briefings in Bioinformatics*, 26(4), 2025.
<https://doi.org/10.1093/bib/bbaf431>
1. Colando, S. & Hardin, J. **Philosophy within Data Science Ethics Courses**, *Journal of Statistics & Data Science Education*, 32(4), 2024. <https://doi.org/10.1080/26939169.2024.2394542>

• In progress

2. Colando, S., Franke, E., Reinhart, A. **Analyzing Statistics Students' Writing Before and After the Emergence of Large Language Models**, submitted, 2026.
<https://doi.org/10.48550/arXiv.2606.22735>
1. Colando, S., Niezink, N.M.D, Jaspers, E. **Why Do Students Avoid Each Other? Investigating Positive and Negative Network Effects**, in progress.

TEACHING EXPERIENCE

Carnegie Mellon Department of Statistics & Data Science

Instructor

Summer Undergraduate Research Experience in Statistics: Sports Analytics Summer 2026
Camp (CMSACamp): website and showcase

Graduate Teaching Assistant

36-402: Undergraduate Advanced Data Analysis	Spring 2026
36-707: Regression Analysis, <i>guest lectured on penalized regression</i>	Fall 2025
Summer Undergraduate Research Experience in Statistics: Sports Analytics Camp (CMSACamp), <i>co-taught four lectures</i>	Summer 2025
36-226: Introduction to Statistical Inference, <i>Head TA</i>	Spring 2025
36-600: Overview of Statistical Learning and Models	Fall 2024

Pomona College

Course Mentor and Grader

MATH158: Statistical Linear Models	Spring 2024
MATH058B: Introduction to Biostatistics	Spring 2023
ID001: First-Year Interdisciplinary Seminar: I Disagree	Fall 2023
BIOL040: Introductory Genetics with Lab	Fall 2022
MATH060: Linear Algebra	Fall 2021
1-2-1 Summer Bridge Program	Summer 2021

PRESENTATIONS

Franke E., Colando, S. "Implications of LLMs on Writing in Statistics and Data Science Courses." Contributed session, International Conference on Teaching Statistics (ICOTS).	Jul. 2026
Colando, S., Franke, E. "Analyzing Students' Statistics Writing Before and After the Emergence of Large Language Models." Main topic session, International Conference on Teaching Statistics (ICOTS).	Jul. 2026
Colando, S., Franke, E. "Analyzing Students' Statistics Writing Before and After the Emergence of Large Language Models." Breakout session, Electronic Conference on Teaching Statistics (ECOTS).	Jun. 2026
Colando, S., Franke, E. "An Introduction to GitHub Classroom." StatBytes, Carnegie Mellon University.	Apr. 2026
Colando, S. "Why Do Students Avoid Each Other? Investigating Positive and Negative Network Effects." Guest lecture for 36-311: Statistical Analysis of Networks, Carnegie Mellon University.	Apr. 2026
Colando, S. "Why Do Students Avoid Each Other? Investigating Positive and Negative Network Effects." ADA Presentation. Carnegie Mellon University.	Nov. 2025
Colando, S. "Why Do Students Avoid Each Other? Investigating Positive and Negative Network Effects." Workshop. Carnegie Mellon University.	Nov. 2025

Colando, S. "Why Do Students Avoid Each Other? Investigating Positive and Negative Network Effects." ICSD Works-in-Progress Seminar. Carnegie Mellon University.	Nov. 2025
Colando, S. "Predicting Avoidance Ties within Social Networks." Networkshop. Carnegie Mellon University.	Mar. 2025
Franke, E. & Colando, S. "Computational Efficiency of R's <code>data.table</code> Package." StatBytes. Carnegie Mellon University.	Mar. 2025
Colando, S. & Hardin, J. "Philosophy Within Data Science Ethics Courses." CAUSE Webinar Series.	Oct. 2024
Colando, S., Pipping, J., Wilson, K. "Clustering Race Horse Movement Profiles to Discover Trends in Injured Horses." Carnegie Mellon Sports Analytics Conference.	Nov. 2023
Colando, S., Pipping, J., Wilson, K. "Clustering Race Horse Movement Profiles to Discover Trends in Injured Horses." Summer Undergraduate Research Experience Showcase. Carnegie Mellon University.	Jul. 2023
Colando, S. "Pomona Funded Summer Experiences Informational Session." Panelist. Pomona College Family Weekend.	Oct. 2022
Chu, C., Colando, S., Nandi, D., Serrano, X. "Quantifying Uncertainty in a Tumor Segmentation Model." A Symposium on Big Data, Human Health, and Statistics. University of Michigan School of Public Health.	Jul. 2022

POSTERS

Colando, S. & Franke E. "Analyzing Statistics Students' Writing Before and After the Emergence of Large Language Models." Research Satellite. United States Conference on Teaching Statistics (USCOTS).	Jul. 2025
Franke, E. & Colando, S. "Teaching Data Cleaning and Wrangling with R's <code>data.table</code> Package." Posters and Beyond. United States Conference on Teaching Statistics (USCOTS).	Jul. 2025
Colando, S. "Analyzing Data Science Ethics Pedagogies." Claremont Center for Mathematical Sciences.	Sep. 2023
Colando, S. "Analyzing Data Science Ethics Pedagogies." Intensive Summer Experience Symposium. Pomona College.	Sep. 2023
Colando, S. "Analyzing ChIP-Seq Normalization Techniques through the Lens of their Biological Assumptions." Intensive Summer Experience Symposium. Pomona College.	Sep. 2022

GRANTS

Carnegie Mellon University Institute of Complex Social Dynamics	2025-2027
Title: "Graduate Student Fellowship"	
Role: Recipient	\$6,000/year
Carnegie Mellon University Department of Statistics & Data Science	2025
Title: "StatBytes Travel Grant"	
Role: Co-recipient	\$500
National Science Foundation (Award Abstract: #2303612)	2025
Title: "data.table Ambassador Travel Grant"	
Role: Co-recipient	\$2,700
Pomona College Department of Mathematics and Statistics	2024
Title: "Kenneth Cooke Summer Research Fellowship"	
Role: Recipient	\$4,200
Pomona College	2022, 2023
Title: "Summer Undergraduate Research Program (SURP)"	
Role: Recipient	\$4,200

LEADERSHIP AND SERVICE

Professional Service

<i>Journal of Statistics and Data Science Education</i> referee	2025-present
CMSAConference Reproducible Research Competition judge	2025
Undergraduate Statistics Project Competition (USPROC) judge	2025
ASA K-12 Data Visualization Poster Competition judge	2025

Carnegie Mellon University Department of Statistics & Data Science

Co-head of Mentorship Committee	2025-present
Co-founder of TidyTuesday Club for undergraduate students	2025-present
Mentor for undergraduate and masters-level students	2025-present

Pomona College Department of Philosophy

Student representative on tenure-track faculty hiring committee	2023-2024
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Pomona-Pitzer Athletics

Varsity women's cross country team captain	2023
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WORKSHOPS AND REU PROGRAMS

Intelligence, Data, Ethics, And Society Summer Program, University of California, San Diego	Summer 2023
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Summer Undergraduate Research Experience in Statistics: Sports Analytics Camp (CMSACamp)	Summer 2023
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Big Data Summer Institute, University of Michigan	Summer 2022
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HONORS AND AWARDS

Passed First-year Ph.D. Data Analysis Exam with Distinction	2025
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Blair Nixon Award	2024
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Bruce Jay Levi Senior Prize	2024
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W.T. Jones Prize in Philosophy	2024
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Phi Beta Kappa	2024
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Sigma Xi	2024
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Pomona College Scholar	2020-2024
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Women's Cross Country and Track and Field Varsity Letter Recipient	2020-2024
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